

Problem Set 6: Photosynthesis

Due: Sunday, November 15 at 11:59 PM

Value: 25 points

Format: Five-question problem set

Purpose

Relate light capture, electron transport, chemiosmosis, carbon fixation, and environmental constraints.

Your task

- Q1 (5): Trace one electron from water to NADPH and name the two photosystems involved.
- Q2 (5): Explain how the thylakoid proton gradient is generated and used, including compartment direction.
- Q3 (5): Predict immediate effects of blocking photosystem II on oxygen evolution, NADPH, and sugar production.
- Q4 (5): Compare cyclic and noncyclic electron flow by inputs, products, and functional value.
- Q5 (5): Interpret the light-response data and distinguish light limitation from carbon or enzyme limitation.

Provided evidence or data

Light intensity	Net O2 production
0	-2.1
100	3.8
300	9.7
600	13.1
900	13.4

Submit

- One PDF with a compartment-labeled electron-flow diagram.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.